

BUS 104: Solving the Diet Problem with Google Sheets and Open Solver

BUS 104: Decision Analysis and Management Science

Danko Turcic

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Welcome to BUS 104 Adventure!

- Instructor: Danko Turcic, Operations Management Guru
- Time: MW 11:00 AM - 12:20 PM
- GitHub Repo: <https://github.com/dxt48/BUS-104>
- Case Pack: <https://hbsp.harvard.edu/import/1456741>
- Goal: Master optimization like a pro!
- Today: Tackle the McDonald's Diet Problem with Google Sheets + Open Solver.
- Engage: Grab your laptop – let's optimize your lunch!

Why Optimize Your Diet?

- Imagine eating McDonald's daily but staying healthy!
- Goal: Meet nutrition needs (vitamins, protein) under 2000 calories.
- Real-world skill: Businesses use this for cost-effective menus.
- Fun Fact: You're learning tools UCR researchers use!
- Challenge: Can you beat the calorie limit with tasty choices?

McDonald's Diet Problem

You want your daily diet plan to have at least 100 percent of the U.S. FDA of vitamin C, calcium; at least 50 grams of protein; and at most 2000 calories. You are wondering if this can be accomplished by eating McDonald's daily nutrition standard.

Menu Item	Price (\$)	McDonald's Food Facts			
		Cal	Protein	Vit. C	Calcium
Hamburger	0.69	260	13	2	15
Big Mac	1.99	560	25	2	25
Chicken Nuggets	1.99	250	15	2	2
Garden Salad	2.05	35	2	40	4
Apple Pie	0.79	260	3	40	2

* Prices recorded Jan. 12, 2000 in St. Louis, Missouri

Step 1: Set Up Google Sheets

- Open Google Sheets (free at sheets.google.com).
- Create a new spreadsheet: Name it “DietProblem2025”.
- Add headers: Menu Item, Price, Calories, Protein, Vit C, Calcium.
- Enter data from the table – let’s build our optimization playground!
- Engage: Type fast – race your neighbor to finish!

Step 2: Install Open Solver

- Open Solver: A free tool for optimization in Google Sheets.
- Go to Extensions > Add-ons > Get add-ons.
- Search “Open Solver”, install it (takes 1 minute).
- Activate: Extensions > Open Solver > Start.
- Fun Tip: Imagine you’re unlocking a superpower for your spreadsheet!

Step 3: Define Decision Variables

- Decision: How many of each item to eat (e.g., 0.5 Hamburger, 1 Salad).
- Add a column: “Quantity” next to each item.
- Enter 0s initially (we’ll let Open Solver decide).
- Why? These are what we’re optimizing!
- Engage: Guess your ideal combo before we optimize!

Step 4: Set the Objective

- Goal: Minimize cost (since prices are given).
- In a cell (e.g., B10), use SUMPRODUCT:
 - Formula: $\text{=SUMPRODUCT}(B2:B6, F2:F6)$ (Price $\times$ Quantity).
 - Label it “Total Cost”.
- Open Solver: Set “Minimize” and select B10 as the target cell.
- Engage: What’s the cheapest healthy meal you can imagine?

Step 5: Add Constraints - Calories

- Constraint: Total calories ≤ 2000 .
- Calculate total calories: $=\text{SUMPRODUCT}(C2:C6, F2:F6)$ (Cal $\times$ Quantity).
- In Open Solver, add constraint:
 - Cell: Calories total cell.
 - ≤ 2000 .
- Fun Check: Will your favorite item fit?

Step 6: Add Constraints - Protein

- Constraint: Total protein ≥ 50 g.
- Calculate: `=SUMPRODUCT(D2:D6, F2:F6)` (Protein $\times$ Quantity).
- In Open Solver, add constraint:
 - Cell: Protein total cell.
 - ≥ 50 .
- Engage: How much protein do you need daily?

Step 7: Add Constraints - Vitamin C

- Constraint: Total Vitamin C ≥ 100
- Calculate: =SUMPRODUCT(E2:E6, F2:F6) (Vit C $\times$ Quantity).
- In Open Solver, add constraint:
 - Cell: Vit C total cell.
 - ≥ 100 .
- Fun Fact: Vitamin C boosts your brainpower!

Step 8: Add Constraints - Calcium

- Constraint: Total Calcium ≥ 10
- Calculate: =SUMPRODUCT(F2:F6, F2:F6) (Calcium $\times$ Quantity).
- In Open Solver, add constraint:
 - Cell: Calcium total cell.
 - ≥ 100 .
- Engage: Strong bones = strong grades!

Step 9: Set Variable Bounds

- Quantities ≥ 0 (can't eat negative food!).
- In Open Solver, add constraint:
 - Cells: Quantity column (F2:F6).
 - ≥ 0 .
- Optional: Limit to whole numbers (e.g., 1 Hamburger, not 0.5).

Step 10: Choose Solver Engine

- Select “Simplex LP” in Open Solver (linear problem).
- Why? It handles linear constraints perfectly.
- Click “Solve” – hold your breath!
- Engage: Predict the result – Salad or Fries?

Step 11: Analyze the Solution

- Open Solver gives optimal quantities.
- Check: Total cost, calories, protein, vitamins.
- Example: 0.5 Garden Salad + 0.1 Hamburger might work.
- Fun Check: Does it meet all constraints?

Step 12: Interpret Results

- If feasible: Celebrate – you've got a healthy, low-cost meal!
- If infeasible: Adjust constraints (e.g., relax calorie limit).
- Real-World: Companies tweak models like this daily.
- Engage: Share your solution with a friend!

Step 13: Sensitivity Analysis

- Why are we eating no Big Macs?
- Removing which constraint would possibly make the optimal menu selection cheaper?
- What if calorie limit changes to 1900?
- What if calorie limit changes to 1500?

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Exercise: Add a Iron Constraint

Consider the same requirement as in the previous example, but require 100% of Iron consumption as well.

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		Cal	Protein	Vit. C	Calcium	Iron
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Big Mac	1.99	560	25	2	25	25
Chicken Nuggets	1.99	250	15	2	2	4
Garden Salad	2.05	35	2	40	4	6
Apple Pie	0.79	260	3	40	2	6

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Step 14: Visualize Your Diet

- Create a pie chart: Select quantities, Insert > Chart.
- Label: Hamburger, Salad, etc.
- Engage: Which item dominates your diet?
- Pro Tip: Share your chart on social media!

Step 15: Troubleshooting Tips

- What if the calorie limit is restricted to 500?
- Error? Check formulas or constraints.
- No solution? Constraints might be too tight.
- Ask in class – we're a team!
- Fun Fix: Pretend you're debugging a video game!

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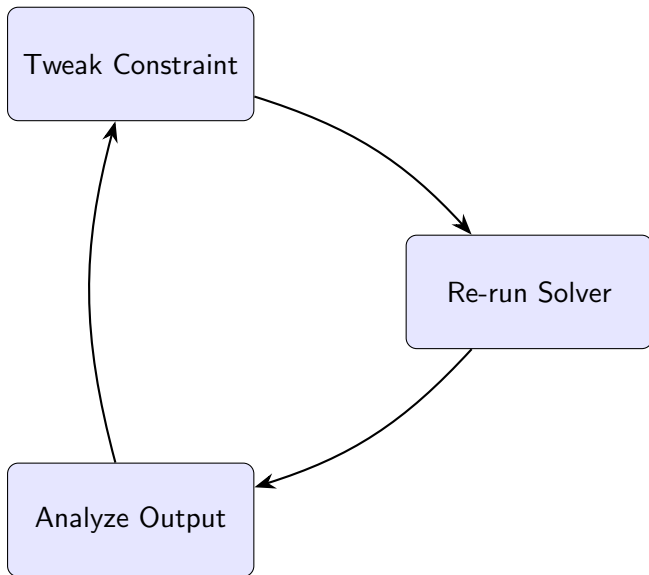
Step 16: Save Your Work

- File > Save (or Ctrl+S).
- Share with classmates via Google Drive.
- Backup: Download as .xlsx.
- Engage: Name your file creatively (e.g., "DietMaster2025")!

Step 18: Real-World Applications

- Airlines optimize fuel with similar tools.
- Hospitals plan patient meals.
- UCR research: Sustainable food supply chains.
- Engage: Where else could you use this?

Bonus Level - The "What-If" Game (Sensitivity Analysis)



Scenario 1 - The Protein Bulking Phase

Protein Constraint:

50g $\longrightarrow$ **150g**

Total Daily Cost?
(Up, Down, Same)

Aggressive Additions?
(Which menu items)

Solving Scenario 1 - The Results

Before (50g Protein)

- High carbs
- Lower total cost
- Cheap fillers

After (150g Protein)

- High protein
- Higher total cost
- McDoubles added

Takeaways:

Tighter constraints = Higher costs

Cheap fillers → Cost-effective protein

New Constraint:

$$\text{Max Calories} \leq \mathbf{1,500 \text{ kcal}}$$

Mathematically possible?

Shift to nutrient-dense?

INFEASIBLE

No mathematical combination satisfies all rules.

The Reality

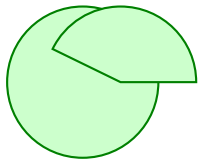
You can't have it all.

The Solution

Relax constraints (spend more, fewer vitamins).

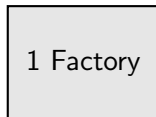
A New Challenge - Integer Linear Programming (ILP)

Continuous Variables

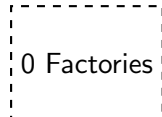


1.43 Salads

Binary (0/1) Variables



OR



0-1 Knapsack Problem

Goal: Maximize Survival Value

Limit: 10 lbs Maximum

Meet Your Gear (The Data)

Item	Value	Weight (lbs)
Tent	10	5
Stove	40	4
Water Filter	30	6
Sleeping Bag	50	7

Translating the Backpack into Google Sheets

	A	B	C	D
1	Item	Value	Weight	Pack It?
2	Tent	10	5	0
3	Stove	40	4	0
4	Water Filter	30	6	0
5	Sleeping Bag	50	7	0
7	Totals	=SUMPRODUCT(B2:B5, D2:D5)	=SUMPRODUCT(C2:C5, D2:D5)	

Solving with OpenSolver (The Magic of `bin`)

- 1 **Objective:** Maximize Total Value
- 2 **Constraint:** Total Weight ≤ 10
- 3 **Variables:** `bin` (Binary 0 or 1)

Optimal Solution:

Stove (4 lbs) + Water Filter (6 lbs) = 10 lbs (Value: 70)

Next Class Preview

- Python optimization next week!
- Bring your curiosity.
- Homework due: Practice this problem.
- Engage: Prep a question for next time!